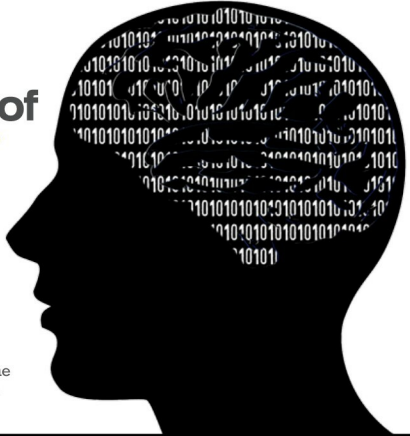


The Importance of In-memory Databases

It has been predicted that in-memory computing will be one of the Top 10 technologies of 2012. In-memory databases (IMDBs) are a critical part of this paradigm. Through this introductory article, let's get acquainted with the basics of IMDBs. We will look at what they are, why they are developed, and the key differences between IMDB and traditional disk DBs.



Gartner has predicted in-memory computing to be one of the top 10 strategic technologies of 2012. In-memory computing is expected to have a disruptive impact on the data warehousing domain in the coming two years. There have been a series of such products released in the last two years, one of the most famous being SAP's HANA. Real-time analytics and sub-second response times for enterprise applications require high-performance data management systems. This, in turn, has led to a surge in the importance of IMDBs.

Network bandwidth has increased dramatically and multi-core processors are available even in mobile phones. However, disk I/O speed has not been increasing at the same rate, which has crippled traditional databases. The first step towards high-performance databases is IMDBs. With ever-growing RAM size and the ability to address more RAM (with 64-bit address spaces), IMDBs are in vogue.

What is an in-memory database?

Wikipedia defines an in-memory database as, "An in-memory database (IMDB; also, main memory database system or MMDB) is a database management system that primarily relies on main memory for computer data storage. It is

contrasted with database management systems that employ a disk storage mechanism."

Margaret H Eich has given a very simple definition of an IMDB/MMDB: a database whose primary data store is main memory.

IMDBs are architected and designed differently from traditional RDBMSs. They have simplified algorithms and mechanisms, which are built with the awareness that all data is going to be in RAM. Figure 1 is a simple illustration of the differences.

IMDBs are also different from simplified storage algorithms like hash-tables or trees, in that IMDBs are 'databases' or RDBMSs. Most IMDBs are ACID-compliant relational databases offering SQL. They have all the properties of a traditional RDBMS, but are tuned for data to be in-memory.

Getting deeper into the IMDB

IMDBs are not a new phenomenon. The great DB scientist Jim Gray conceptualised this technology 30 years ago and also predicted that in the 2000s these technologies would become widely accepted. Almost from the early 1980s, IMDBs have existed in the telecom and defence domains. However, these